

**288R Project Progress Report - Yelp-Powered Insights: Building a Business
Recommender System**

Project Group #: 07

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1. Background

a. 2 lines on a problem that is planning to solve. Why is it important to solve?

Users often have to scroll and read through multiple reviews before landing on the business that meets their preferences and needs. Clicking into each business and reading the reviews is time-consuming for the user and too many choices for the user can become overwhelming. This project aims to create a recommendation algorithm to help users discover businesses that match their preferences by analyzing user interaction data, business details, friends, and geographic location. The recommendation algorithm will solve the problem by giving personalized recommendations, increasing awareness of businesses that they might not have otherwise considered, and helping make the decision-making process quicker.

2. Dataset name and citation/link

Yelp Complete Open Dataset

<https://www.kaggle.com/datasets/adamamer2001/yelp-complete-open-dataset-2024>

3. Data Pipeline

a. Purpose

b. Pipeline design and details

c. Outputs

The purpose of this pipeline is to efficiently create a streamlined approach to filter, explore, and investigate the information that is needed to create the recommendation algorithm.

Data Transformations

Step	Purpose	Input	Prior Step	Output	Tool/Library
1	Combine dataframes	Reviews DF Buisness DF	Obtaining data from Kaggle and loading	Combined DF	Pandas

			as df		
2	Rename columns	Combined DF	Combine dataframes	Renamed DF	Pandas
3	Drop unnecessary columns	Renamed DF	Rename columns	DF with necessary columns	Pandas
4	Drop rows with missing values	DF with necessary columns	Drop unnecessary columns	DF with no missing values	Pandas
5	Aggregate average user rating	Reviews DF	Obtaining data from Kaggle and loading as df	Average ratings DF	Pandas
6	Add <i>avg_stars</i> column	DF with no missing values & Average ratings DF	Drop rows with missing values & Aggregate average user rating	DF with average ratings	Pandas
7	Aggregate number of friends	User DF	Obtaining data from Kaggle and loading as df	User DF with an added column	Pandas
8	Add <i>num_friends</i> column	DF with average ratings & User DF with an added column	Add average ratings column & Aggregate number of friends	DF with the number of friends	Pandas

Our pipeline processes and analyzes Yelp's business, user, and review data to generate a personalized business recommendation list while ensuring data cleanliness, consistency, and readiness for machine learning models. It consists of six key stages: data extraction, data cleaning and preprocessing, exploratory data analysis (EDA), feature engineering, model training, and validation.

- Data Extraction:

We extracted five JSON files from Kaggle, imported them into Python, and converted them into data frames for further processing. To obtain a final data frame that includes all the columns needed to build the model, we combined review data with business data by using the common column business ID to merge the datasets.

- Data Cleaning & Preprocessing:

We examined missing values (37,070 entries) and 0 duplicated records, addressing these issues by removing incomplete and redundant data to ensure data integrity. Data formats did not have an issue but column types were corrected such as changing Object types to string or booleans. Columns were also renamed for clarity such as specifying whether the star rating is for reviews or businesses. Data normalization was not shown to be necessary at this current step therefore no columns were normalized.

- **Exploratory Data Analysis (EDA):**

We explored its structure to gain insights into the data, including column distributions, variable types, and key statistics. The data was pretty complete and high quality after merging. All required columns for building the model were present. There were no outliers except for the column number of friends. There was no missing data, and data formats and naming conventions were taken care of.

- **Feature Engineering:**

We extracted and transformed key features for model training, ensuring they aligned with machine learning models' assumptions. Additional attributes include aggregating average user ratings and the number of friends in the user data. After the aggregation step, we further merged the combined DF with user data using user ID to add the additional features of average user ratings and number of friends to the merged DF.

- **Model Training & Validation:**

The data will be split into 80% training, 10% validation, and 10% test sets, this is a common baseline that works well with large datasets. We will implement a hybrid recommendation approach, combining collaborative filtering and content-based methods, using ML algorithms such as KNN and Random Forest to enhance accuracy. Additional techniques like cosine similarity, Jaccard similarity, and collaborative filtering will be applied to refine recommendations.

- **Evaluation:**

We will assess model performance using accuracy, precision, recall, and F1 score, ensuring the effectiveness of the generated recommendation list.

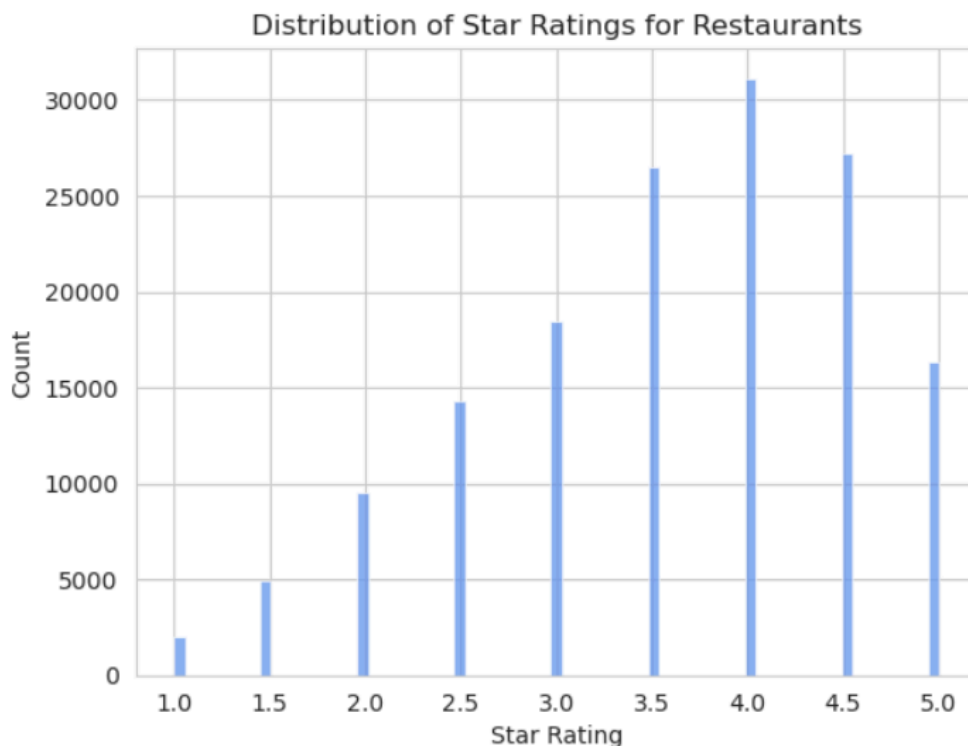
The expected output consists of three key stages: (1) a clean, deduplicated, and structured dataset ready for model training, (2) user-specific business recommendations generated based on past interactions, reviews, and locations, and (3) an evaluation of the recommendation list's quality using performance metrics.

4. EDA Description

- a. Types of analysis used (Univariate, Multi-variate, Graphical/Non-Graphical)**
- b. Artifacts produced**

We conducted exploratory data analysis (EDA) to gain insights into the dataset's volume, data types, and overall distribution. Our analysis included both univariate and multivariate methods, utilizing graphical and non-graphical techniques to uncover key patterns and relationships.

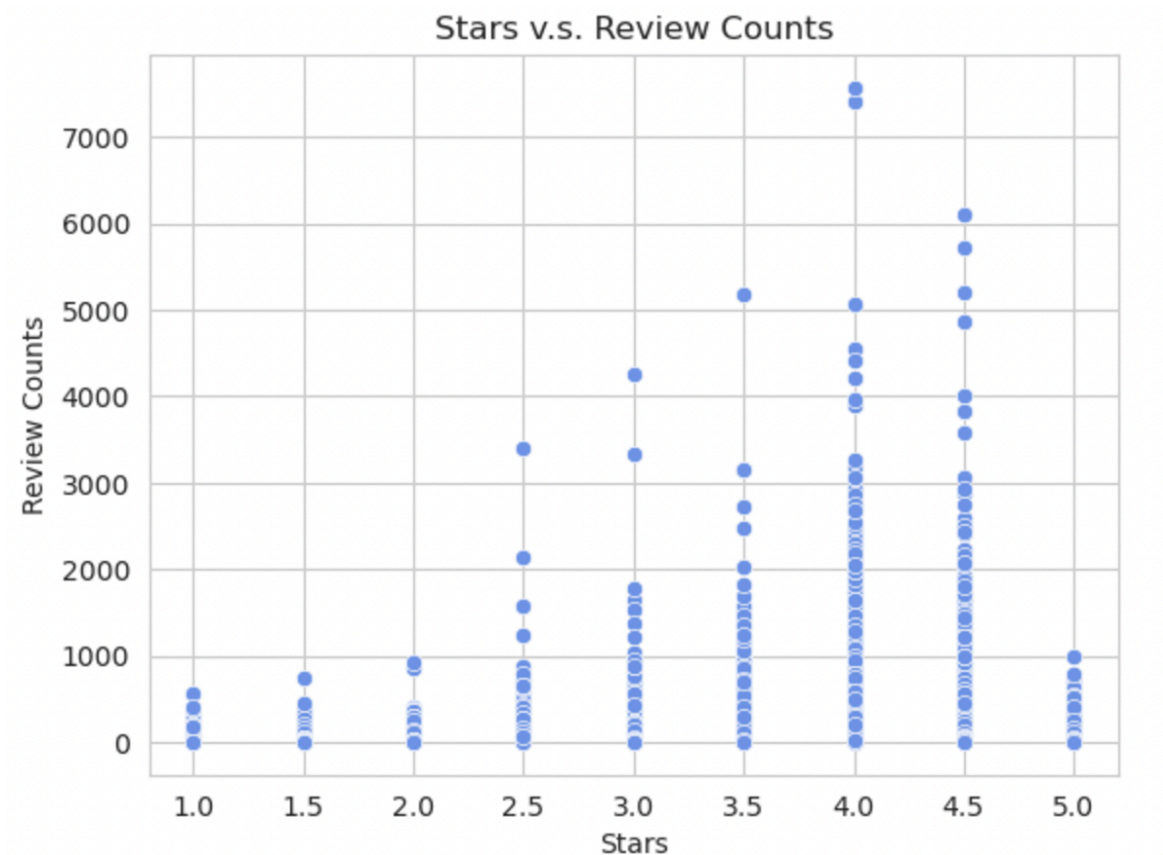
For univariate analysis, we examined the distribution of variables such as business ratings and review counts. For instance, we found that the dataset contains 150,346 businesses, with the number of reviews per business ranging from 5 to 7,568. By plotting the distribution of business star ratings with a frequency bar, we observed that the majority of businesses received a rating of 4, while about 2,000 businesses had the lowest rating of 1.



We also plotted the distribution of review star ratings with a frequency bar which shows that most reviews give the highest rating of 5 stars and the chance of giving higher ratings gradually increases from giving 2 stars to giving 4 stars with 1 star rating being the third highest.



For multivariate analysis, we explored relationships between different variables. For example, we analyzed the correlation between review count and star rating and discovered an interesting pattern: businesses with ratings above 4 and below 2 tend to have higher review counts, while 5-star ratings had the most reviews overall. This suggests that customers are more likely to leave reviews when they are highly satisfied or deeply dissatisfied with a business, which further proves the previous plot in review star rating distribution.

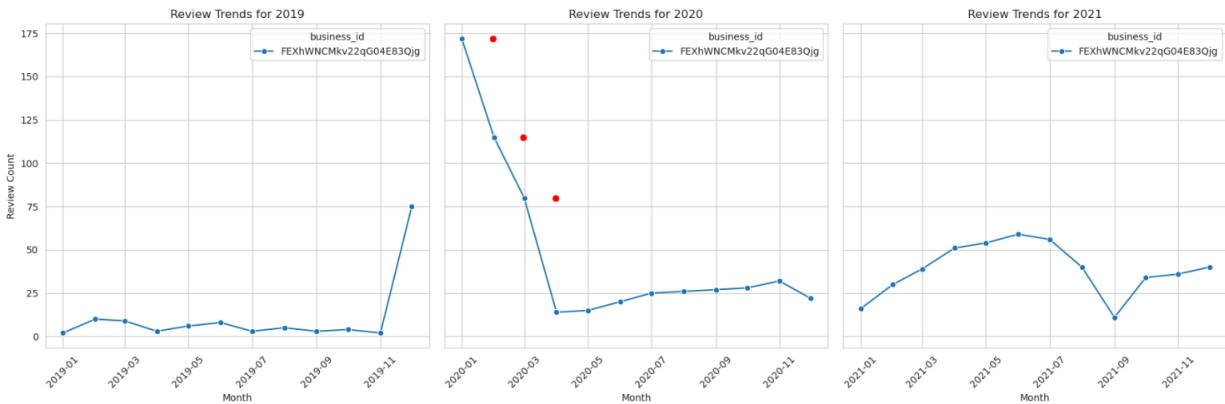


We took a look at business categories and saw that for the categories below they got the highest scores. Note: Business can be classified as more than one category. These insights could help to understand how to modify in case these categories might not have a lot of reviews that could give good recommendations to users.

	category	stars
171	Bubble Soccer	5.000000
1116	Sport Equipment Hire	5.000000
868	Patent Law	5.000000
767	Mohels	5.000000
750	Metal Detector Services	5.000000
660	Karaoke Rental	5.000000
241	Childproofing	4.975000
235	Cheese Tasting Classes	4.941176
1229	Undersea/Hyperbaric Medicine	4.938462
194	Calligraphy	4.937500

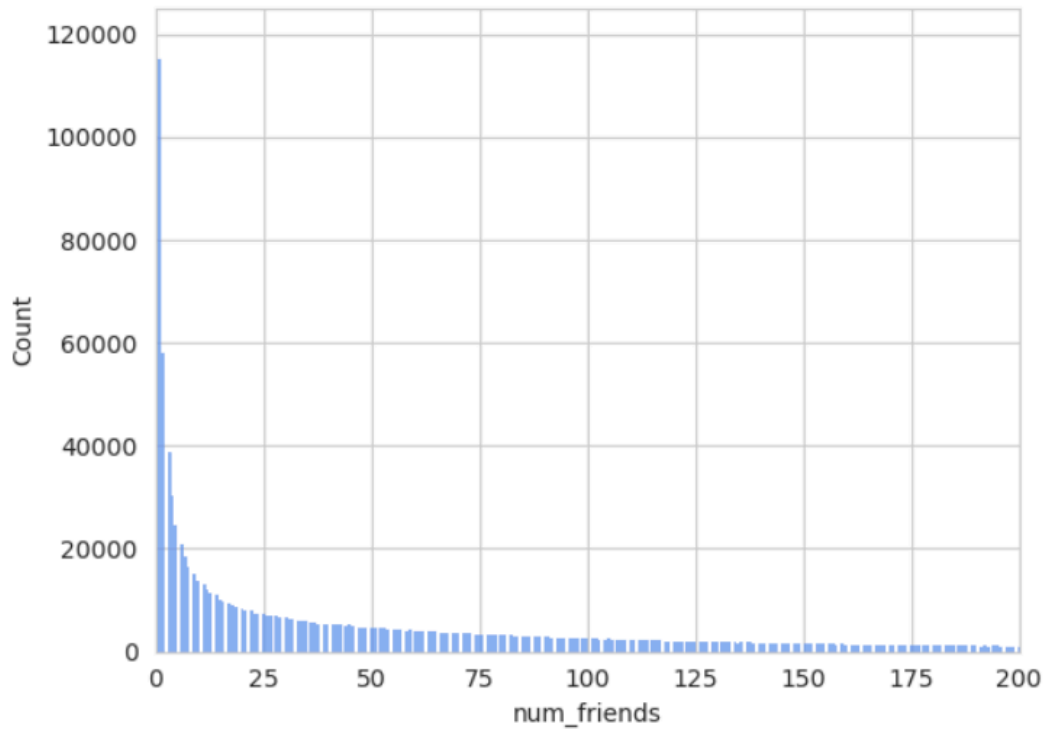
We also took a look at businesses that had higher than-normal reviews. The below charts show for one business the red dots are the abnormal number of reviews that it

received, so in 2020 from January to March, there were more reviews for this business than what it normally received. When building a recommender system, we want to take into consideration that different months and years could be more popular for a business. Maybe the business attracted more publicity etc.

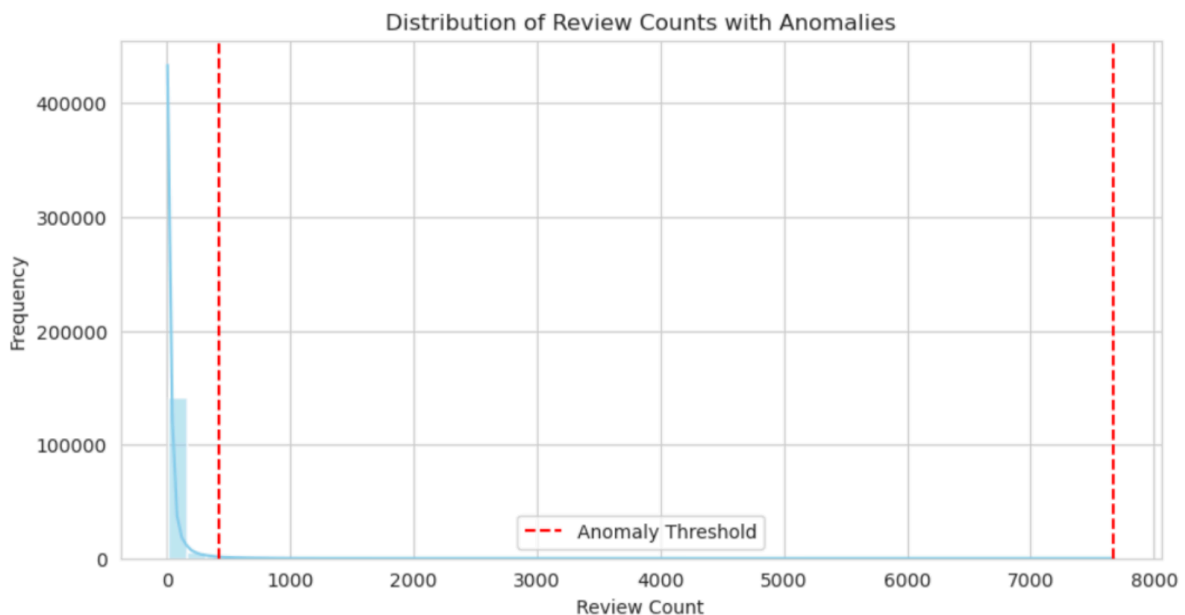


These insights from EDA helped us better understand the dataset's structure and guided our feature engineering and model selection for the recommendation system.

For the outliers/anomalies, some people have a lot more friends than others this data can be useful to understand patterns and see if having more friends gives more recommendations than people who have fewer friends.



We also took a look at review counts that were a lot higher for each business as shown there were anomalies with this as well. Some businesses had fewer review counts compared to other businesses. Took the $\text{abs}(\text{z-score}) < 3$ to find these anomalies.



5. Feature Engineering

- For one more or more of the planned models?

**b. If yes, what features do you plan to extract? Used in any literature?
Other features you intend to use and how you came up with them.
Start from EDA and trace a link to the feature**

To maximize the performance of our recommendation system, we plan to engineer features that capture key aspects of user preferences, business attributes, and review patterns. Based on our EDA, we identified key variables and patterns that guided our feature selection. Our feature engineering process includes four main categories.

For user-based features, we aggregate past interactions such as average ratings given by a user and the number of reviews issued to encode user preferences. Examining user engagement patterns helps inform collaborative filtering models.

For business-based features, we will extract business attributes such as food type or service type to improve content-based filtering. Since these attributes are mostly text-based, we plan to apply TF-IDF vectorization to extract meaningful keywords.

For review-based features, we will perform sentiment analysis on user reviews using NLP techniques to quantify customer satisfaction. As well as derive frequency-based sentiment metrics, such as the percentage of positive and negative reviews for each business, to enhance recommendation accuracy.

For similarity metrics, we will compute cosine similarity and Jaccard similarity to measure similarities between users and businesses, refining the collaborative filtering model. Also will be able to understand how the user's friends can also help in refining places the user may not have visited before.

Our EDA played a critical role in shaping our feature selection process. By analyzing missing values in key variables, we ensured that our dataset remained complete and reliable for model training. Examining the correlation between review count and star ratings, we identified that businesses with very high or very low ratings received more reviews. This insight motivated us to include review sentiment as a predictive feature.

Many existing studies have used similar techniques to enhance recommendation systems. For example, Catherine applied sentiment analysis in her restaurant recommender system using TextBlob's Polarity Score to quantify review sentiment and incorporate it as a feature in her model. These engineered features will be utilized in our hybrid recommendation system, where user-based and business-based features will inform content-based filtering models such as Random Forest, while similarity-based

features will refine collaborative filtering models, including KNN and matrix factorization approaches.

6. What models are you using?

- a. List the models in order of highest to lowest priority for implementation**
 - i. Model architecture from literature**
 - ii. New architecture experiences you want to try**

We will implement a hybrid recommendation system, prioritizing models in the following order: collaborative filtering, content-based filtering, and similarity-based methods.

We will prioritize collaborative filtering due to its widespread success in recommendation systems, particularly in platforms like Amazon and YouTube. Collaborative filtering relies on past user interactions with items to recommend new ones, focusing on groups of users with similar preferences rather than individual item features. This approach has proven effective in capturing user behavior patterns and making personalized recommendations without needing item-specific information.

Next, we will incorporate content-based filtering, which recommends items based on their attributes and user preferences. This model complements collaborative filtering by considering item features, such as business categories and descriptions, to suggest businesses that align with user interests.

Finally, we will use similarity-based methods, such as cosine similarity and Jaccard similarity, to refine both collaborative filtering and content-based methods. These techniques measure the proximity between users and businesses, further enhancing the accuracy of the recommendations.

Collaborative filtering is widely recognized in literature and serves as the foundation for many successful recommendation systems. One potential area we want to explore is enhancing content-based filtering by integrating sentiment analysis, providing more nuanced recommendations based on review sentiments. Additionally, we plan to experiment with tuning collaborative filtering algorithms to optimize their accuracy and diversity in recommendations.

7. Progress Report Details on your progress

- a. The effort is taken on dataset collection, feature extraction, what tests you have run**

For the milestone 2 report, we focused on dataset collection, feature extraction, and testing to ensure the data's quality and relevance for our recommendation system.

We collected the Yelp Complete Open Dataset from Kaggle, which includes data on businesses, reviews, and user interactions, providing the foundation for analyzing user preferences. For feature extraction, we aggregated user interactions like average ratings and the number of reviews to capture preferences. We planned to use TF-IDF vectorization to extract business attributes such as food and service types and apply NLP techniques for sentiment analysis, extracting positive and negative review percentages. We performed tests by checking for missing values and duplicates and handling them appropriately. EDA helped us understand feature distributions and correlations, guiding feature selection.

8. Team member contribution

a. What is every member working on/planning to work on?

For milestone 2, responsibilities are divided as follows: Tim was responsible for uploading the data and initiating and refining the EDA and report. Claire and Cassy collaborated with the TA to resolve technical issues, refined the EDA, and contributed to finalizing the report. The tasks are evenly distributed among the three of us to ensure efficient progress.

9. Risks

a. List any risks you see at this stage to completing the project you defined in the abstract and how you are mitigating them

At this stage, there are two risks that need to be addressed. The first risk is the presence of missing values in our features, which could affect the accuracy of our recommendation system. To mitigate this, we have removed missing and duplicate values to ensure dataset consistency for further analysis. The second risk is related to the size of our dataset, as working with large data and complex algorithms has caused a few crashes during execution. To manage this, we have decided to retain only the relevant features, reducing the data size for faster execution and more efficient training.

10. References

Gitau, Catherine. "Restaurant Recommender System". Jan 26, 2021.

<https://categitau.medium.com/restaurant-recommender-system-eefd3226ed61>

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